

The Ecuadorian Electricity Market: A Hybrid Model of Sovereignty and Strategic Regulation

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Abstract—This article examines the legal viability and operational structure of the electricity market in Ecuador, with particular attention to the tension between the constitutional mandate of state sovereignty and the need for private investment in a strategic sector. The study is based on a systematic analysis of the 2008 Constitution, the 2015 Organic Law of the Public Electricity Service (LOSPEE), and the 2024 Organic Law on Energy Competitiveness (LOCE). It concludes that the Ecuadorian system does not constitute a liberalized market, but rather a hybrid model of regulated expansion. Exceptional delegation has enabled competitive contracting mechanisms, while the LOCE has broadened private participation through private-initiative projects, transmission concessions, and a distributed-generation micro-market. The article characterizes the structure of this market and demonstrates that state control is preserved primarily through centralized planning and tariff regulation.

Index Terms—Electricity market, exceptional delegation, LOSPEE, LOCE, public selection process (PPS), regulated tariffs.

I. INTRODUCTION

THE Ecuadorian electricity sector is frequently characterized as a vertically integrated state monopoly, a perception grounded in the robust constitutional framework that designates energy as a strategic sector [1]. A closer legal analysis, however, reveals an institutional design that not only permits but increasingly enables and encourages the use of market mechanisms. The central legal question is therefore not whether a market exists, but what type of market Ecuadorian law permits.

Ecuador's legal framework does not establish a liberalized market characterized by open access and freely determined spot prices. Instead, it provides for a hybrid market undergoing deliberate expansion. This market has two principal components: regulated competition for entry into the generation segment and, more recently, the transmission segment; and an emerging self-consumption “micro-market” for prosumers, understood as consumers who also produce electricity. Both components remain subject to strict state direction and oversight, particularly with respect to the determination of final consumer tariffs.

II. PRELIMINARY DEFINITIONS

An assessment of the scope and structure of Ecuador's electricity sector, as well as its increasing use of market mechanisms, requires clarification of the principal legal and technical concepts underlying the analysis.

This paper was developed by the Distribution Networks group of the Electrical Engineering program at Universidad Nacional de Loja.

A. The State

In public law and political economy, the State is not merely the governmental apparatus; it is the sovereign political organization of society. For the purposes of this analysis, the State is understood in accordance with the 2008 Constitution, which defines Ecuador as a “constitutional State of rights and justice” [2].

The State performs a dual role in the electricity sector. First, it exercises sovereignty over natural resources, including energy resources. Second, it bears the legal responsibility for ensuring the provision of public services. It directs, regulates, and supervises strategic sectors and determines whether, when, and under what conditions the performance of particular activities may be delegated to private entities.

B. Electricity Market

An electricity market is the physical, financial, and regulatory system through which electricity and related services are purchased and sold. Unlike a conventional goods market, an electricity market must continuously balance supply, represented by generation, and demand, represented by consumption, because of the physical limitations associated with large-scale electricity storage [3].

Two principal levels may be distinguished:

- **Wholesale market:** Generators sell large quantities of electricity to distribution companies or large industrial consumers.
- **Retail market:** Suppliers sell electricity to end users.

In Ecuador, the concept of a market must be qualified because the sector operates in a highly regulated environment in which competition generally concerns rights of entry awarded through public selection procedures rather than freely determined prices.

C. Constitution of the Republic

The Constitution is the supreme norm of Ecuador's legal order (*norma normarum*). Under the principle of constitutional supremacy, all statutes, regulations, and other legal provisions must conform to it.

In the electricity sector, the Constitution is more than a general legal framework. It establishes ownership of resources, defines the strategic nature of the sector, and specifies the exceptional circumstances under which private parties may participate. It is therefore the primary legal source that enables and limits the operation of market mechanisms.

D. Organic Law of the Public Electricity Service

The Organic Law of the Public Electricity Service (LOSPEE) is the specialized organic statute that implements the constitutional mandates applicable to electricity. It defines the sector's institutional structure, the rights and obligations of its participants, and the specific mechanisms governing tariffs and sanctions [4].

Whereas the Constitution establishes *what* the State must do—retain control while permitting delegation—the LOSPEE establishes *how* delegation may occur, including the applicable contracts, procedures, and technical and economic conditions. It is thus the operational statute that translates constitutional principles into an industry model.

III. CURRENT SITUATION IN ECUADOR

The question of whether an electricity market exists, or may legally exist, in Ecuador reveals an apparent conflict between state sovereignty and the need for private investment. Analysis of the legal framework, from the Constitution to the most recent reforms considered in this study, demonstrates that such a market is not only legally possible but already operational and expanding.

Nevertheless, Ecuador does not have a liberalized market for final consumer prices. It has a hybrid, regulated model intended to reconcile state control with private-sector efficiency and capital. It is principally a market for investment and contracts rather than a market of freely determined retail tariffs.

A. Constitution of the Republic of Ecuador

The 2008 Constitution of the Republic of Ecuador [2] establishes a robust framework of state sovereignty. Article 313 unequivocally provides that “the State reserves the right to administer, regulate, control, and manage strategic sectors.” It expressly includes “energy in all its forms” among those sectors [1], [2].

Article 314 reinforces this principle by establishing the public-service model and assigning the State responsibility for the provision of electricity as a public service [1]. It requires public services and their provision to comply with the principles of obligation, generality, uniformity, efficiency, responsibility, universality, accessibility, regularity, continuity, and quality. It further requires the State to ensure equitable prices and tariffs and to establish mechanisms for their control and regulation [2].

Together, these provisions establish a presumption that the State is the principal provider and ultimate guarantor of the service, an arrangement that appears, at first sight, inconsistent with freely determined prices and open competition.

Article 316 of the Constitution, read in conjunction with the Organic Code of Production, Trade and Investment (COPCI) [5], resolves this tension by providing a legal basis for market participation [2]. It states:

“The State may delegate participation in strategic sectors and public services to mixed-capital companies [...] and may, on an exceptional basis, delegate the exercise of these activities to private initiative

and the popular and solidarity economy in the cases established by law.”

This provision constitutes the constitutional basis for a market. It reconciles the state control required by Article 313 with the private investment that may be necessary to satisfy the public-service obligations imposed by Article 314.

The existence, nature, and limits of any electricity market in Ecuador consequently depend on how secondary legislation, particularly the LOSPEE, defines the exceptional circumstances and cases in which delegation is permitted.

Such delegation does not transfer ownership of the underlying resource. Article 408 establishes that natural resources and energy are the inalienable, imprescriptible, and non-seizable property of the State. It also requires the State's share of the benefits derived from the exploitation of these resources to be no less than that received by the undertaking that exploits them.

Accordingly, the authorized market is not a market for ownership of the resources themselves. It is a market for the management, technology, and investment required to convert state-owned energy resources into electricity.

Additional constitutional provisions support a mixed and regulated energy sector:

- Article 283 provides that the economic system comprises public, private, mixed, popular and solidarity-based forms of economic organization, as well as other forms recognized by the Constitution. The popular and solidarity economy is regulated by law and includes cooperative, associative, and community sectors.
- Article 284 defines the objectives of economic policy, including the promotion of national production, systemic productivity and competitiveness, the accumulation of scientific and technological knowledge, strategic participation in the global economy, and complementary productive activities within regional integration.
- Article 304 identifies the development, strengthening, and stimulation of domestic markets, in accordance with the strategic objectives of the National Development Plan, as an objective of trade policy.
- Article 315 authorizes the State to establish public enterprises to manage strategic sectors, provide public services, ensure the sustainable use of natural resources or public goods, and undertake other economic activities.
- Article 319 recognizes diverse forms of productive organization, including community, cooperative, public and private business, associative, family, domestic, autonomous, and mixed forms. It therefore requires the State to encourage production that satisfies domestic demand and ensures Ecuador's active participation in the international context.
- Article 335 requires the State to define a pricing policy intended to protect national production and to establish sanctions against private monopolistic and oligopolistic practices, abuse of market dominance, and other forms of unfair competition.
- Article 413 requires the State to promote energy efficiency; environmentally sound practices and technologies; and diversified, low-impact renewable energy that

does not endanger food sovereignty, ecological balance, or the right to water. This provision provides constitutional support for renewable-energy investment, including potential private investment in that subsector.

In summary, although the Constitution does not expressly establish a competitive “electricity market,” it permits exceptional delegation to private entities. It thereby creates limited opportunities for private participation without requiring comprehensive privatization.

Nor does the Constitution categorically prohibit an electricity market or private participation. It recognizes the market as part of the economic system under Article 283, while emphasizing that the State remains responsible for electricity provision under Article 314 and may operate through public enterprises under Article 315. Moreover, Article 313 classifies energy management as strategic, thereby requiring a substantial and continuing state role.

B. Organic Law of the Public Electricity Service

The Organic Law of the Public Electricity Service (LOSPEE), enacted in 2015 [4], implements the “exceptional delegation” authorized by Article 316 of the Constitution. The statute defines the public electricity service as encompassing generation, transmission, distribution, supply, and public lighting. The initial opening to private capital, however, was concentrated primarily in the generation segment.

Article 10 of the LOSPEE expressly recognizes a plurality of participants. It provides that the sector operates through public enterprises, mixed-capital companies, private companies, consortia or associations, popular and solidarity economy enterprises, and foreign state-owned companies.

The admission of private-sector competition responds to the technical and economic challenges identified in the statute itself:

- **Demand growth:** Population and economic growth require additional sources of electricity supply.
- **Energy transition:** The system must develop an economically and environmentally balanced energy mix by reducing inefficient thermal generation and increasing renewable generation, including wind, biomass, and photovoltaic resources.
- **Modernization:** The sector must integrate technologies such as smart grids, active demand management, and distributed generation.

The LOSPEE establishes a clear hierarchy of participation. Article 24 sets forth the general rule that the State, acting through public enterprises, is the principal entity authorized to undertake all sector activities. As a subsidiary mechanism, Article 19 of the LOSPEE Regulations [6] permits mixed-capital companies in which the State holds a majority interest to participate when the public sector cannot develop particular projects.

Article 25 of the LOSPEE, however, establishes the exception that enables market participation. The competent ministry may delegate sector activities to private-capital companies and popular and solidarity economy enterprises in the following circumstances:

- 1) When delegation is necessary to satisfy the public, collective, or general interest.
- 2) When public or mixed-capital companies cannot meet demand for the service.
- 3) When projects use non-conventional renewable energy sources and are not included in the Electricity Master Plan (PME).

The legal framework provides both a competitive mechanism and a legal instrument through which such delegation may be implemented.

Market access is competitive rather than automatic. Article 52 of the LOSPEE and Articles 20 and 21 of its Regulations [6] provide that the competent ministry shall conduct Public Selection Processes (PPS) for priority or delegated projects. These procedures seek to identify the bid or bids offering the most favorable conditions for project development.

Once a project has been awarded through a PPS, the legal relationship is formalized by means of an enabling instrument. Article 29 of the LOSPEE and Article 3 of its Regulations define the **concession contract** as the legal instrument authorizing a private company to construct and operate a generation facility.

These provisions define the essential nature of Ecuador’s electricity market. It is not a spot market based on daily competition for dispatch; rather, it is a long-term procurement market characterized by competition *for* the market rather than competition *within* the market.

Under this arrangement, the State retains direction of the sector through centralized planning under the Electricity Master Plan, which determines the capacity required by the system. It subsequently uses the PPS as a cost-optimization mechanism through which private parties compete for the right to execute a concession contract, generally by offering a fixed electricity sale price. Although this framework does not amount to full market liberalization, the introduction of short-term transactions and bilateral contracts indicates movement toward more dynamic mechanisms and demonstrates that private participation can coexist with state regulation.

C. Organic Law on Energy Competitiveness

Whereas the 2015 LOSPEE opened the generation market to private participation, the Organic Law on Energy Competitiveness (LOCE), published in Official Gazette No. 475 on January 11, 2024 [7], significantly broadened that opening. Prompted by recurring energy crises and the urgent need to accelerate investment [8], the LOCE amended the LOSPEE to extend the scope of the “exceptional delegation” authorized by Article 316 of the Constitution across three critical segments of the electricity supply chain.

The most significant amendment concerns Article 25 of the LOSPEE. The revised provision expressly permits delegation through PPS procedures to private-capital companies and foreign state-owned companies. More importantly, it authorizes non-conventional renewable-energy projects that are not included in the Electricity Master Plan.

This amendment changes the structure of the market:

- **LOSPEE: A reactive market.** The State identified and planned projects, while private entities could bid only for projects requested by the State.
- **LOCE: A proposal-driven market.** Private initiative allows investors to identify and propose projects not previously contemplated by the State, thereby reducing strict dependence on centralized planning.

The LOCE also extends private participation to segments previously treated as strict state or natural monopolies.

Transmission had historically been reserved for CELEC EP. Although it remains primarily a state-provided public service, the LOCE exceptionally authorizes private companies and foreign state-owned companies to construct **new transmission projects** under concession arrangements. This reform creates an investment market for critical infrastructure, including the high-voltage lines required to connect renewable-energy facilities in remote areas to the National Interconnected System (SNI).

The statute also creates a specialized market for network expansion. Infrastructure intended to supply commercial or industrial customers that are **isolated from the grid** may be constructed by those customers or by other interested legal entities selected through a competitive process. Although such assets ultimately revert to the state-owned distribution company, the arrangement allows private entities to compete to finance and implement network expansion.

The enactment of the LOCE in 2024 therefore represents a pragmatic shift in Ecuadorian energy policy, moving the country from a centrally planned market toward a market of incentivized expansion.

The legislation implicitly recognizes that a model based exclusively on state investment is insufficient to satisfy increasing demand. Ecuador's electricity market is consequently a dynamic legal construct whose boundaries across generation, transmission, and distribution are adjusted by the legislature to address public-policy concerns and ensure security of supply.

D. Organic Code of Production, Trade and Investment

Article 98 of the COPCI provides that new electricity-generation projects developed by private Ecuadorian companies shall receive the same treatment, mechanisms, and payment or guarantee conditions for electricity purchases as those applicable to international electricity transactions. Such treatment remains subject to the relevant decisions of the Andean Community, the rules issued by the electricity-sector regulator concerning payment guarantees, and a favorable project-specific report from the ministries responsible for electricity and public finance.

E. Consumer Market

Ecuador's legal framework has created not only a wholesale procurement market through the PPS mechanism, but also a micro-market at the end-user level that enables passive consumers to become prosumers.

The LOCE is again the principal statutory basis. It inserts an unnumbered article after Article 44 of the LOSPEE that permits regulated and unregulated consumers to install

distributed-generation systems using non-conventional renewable energy sources exclusively for self-supply [7].

On this statutory basis, ARCONEL issued Regulation No. ARCONEL-010/24 [9], which establishes the regulatory framework for distributed generation used by regulated electricity consumers for self-supply.

This regulation creates the market mechanism applicable to prosumers by establishing rules for electricity metering and billing.

Academic analyses of the Ecuadorian model characterize the applicable arrangement as net metering [10]. Unlike a feed-in tariff, net metering offsets energy imports and exports on a one-to-one energy basis.

The prosumer effectively uses the distribution grid as a "virtual battery." When the prosumer generates surplus electricity, a bidirectional meter records the corresponding kilowatt-hours (kWh) as a credit. When electricity is drawn from the grid, the relevant kWh are deducted from that credit. At the end of the billing period, the user pays only for the net difference between electricity consumed and electricity injected into the grid [10]. This arrangement constitutes a market transaction in which the consumer's surplus is credited at the same energy rate applied to electricity drawn from the grid.

In parallel with net metering for smaller consumers, the law regulates self-generators, which are generally large industrial users. The LOCE and the associated regulatory amendments have integrated these participants into the wholesale market.

The amendment to the LOSPEE Regulations [11] provides:

"For self-generators with geographically dispersed own-use loads, both the demand associated with those loads and the electricity produced by the self-generation facility shall be considered in economic dispatch."

Unlike a distributed-generation prosumer operating under a net-metering arrangement with a distribution company, a large industrial self-generator participates in national dispatch. CENACE treats its production and demand as components of aggregate system supply and demand, thereby allowing it to participate in the wholesale market and execute regulated contracts.

F. Market Limits

Notwithstanding the clear authorization and expansion of market mechanisms, Ecuadorian legislation imposes strict limits consistent with the constitutional requirement of state direction.

The existence of a generation procurement market based on PPS procedures and a distributed-generation micro-market must not be confused with a liberalized retail-pricing system. Article 314 of the Constitution requires equitable tariffs subject to state control and regulation.

ARCONEL implements this mandate through codified Regulation No. 004/24 [12], which establishes the economic and tariff regime. Its purpose is not price discovery through the interaction of supply and demand, but the establishment of a conceptual and methodological framework for determining costs and setting tariffs.

Tariff regulation is therefore the principal constraint on liberalization. Although generation companies compete for market entry, the final consumer price is not freely determined. ARCONEL sets tariffs through a cost-based methodology intended to preserve the economic balance of the electricity sector and comply with the constitutional mandate.

The PPS procedures established by the LOSPEE and the LOCE are not the only legal mechanism available for private investment. Ecuador also has an Organic Law on Incentives for Public–Private Partnerships (PPP) [13].

The regulations implementing that law confirm its application to contractual delegations of management in strategic-sector projects. Its principal advantage for investors in a regulated market is the assurance of stability in the sector-specific legal framework for the term of the contract.

This framework creates an alternative or complementary mechanism for project development. An investor may structure an energy project under the LOSPEE and LOCE framework or under the PPP regime. The existence of the PPP statute recognizes that, in a market without freely determined prices, attracting capital requires long-term contractual and legal certainty.

The Ecuadorian model is therefore not a single market but a dual system: a regulated-price market for final consumers and an investment-contract market for expanding generation and transmission, in which private entities compete for contracts through either LOSPEE and LOCE selection procedures or the PPP framework.

IV. CONCLUSION

Analysis of Ecuador’s regulatory framework supports an affirmative conclusion: an electricity market is both legally permissible and operationally present in Ecuador. Its characteristics, however, differ substantially from those of conventionally liberalized markets.

On the basis of the Constitution, the LOSPEE, and the 2024 LOCE, Ecuador’s system should be classified neither as a free market nor as a purely state-owned monopoly. It is more accurately described as a *hybrid model of regulated expansion*, arising from the legally managed tension between state direction under Article 313 and exceptional delegation under Article 316.

Within this model, private participation is not an end in itself but a pragmatic instrument. The State uses market mechanisms to attract the investment required to fulfill its binding constitutional obligation to guarantee a continuous, efficient, and high-quality electricity service. The constitutional model therefore does not prohibit the market; it subordinates market activity to state planning, regulation, and oversight.

The current legal framework establishes a market operating at three distinct levels:

- 1) **Competition for the market:** This is the most developed segment. Although the State retains overall planning authority, it uses private competition through PPS procedures to expand generation capacity. The 2024 LOCE strengthened this mechanism by authorizing private-initiative projects outside the Electricity Master

Plan and, for the first time, permitting concessions for new transmission infrastructure.

- 2) **Distributed generation:** The LOCE and ARCONEL regulations have enabled a decentralized market. Through net metering, users become prosumers and exchange electricity with the grid. In parallel, large industrial self-generators participate in national economic dispatch and therefore contribute actively to wholesale-market supply and demand.
- 3) **Consumer protection:** Unlike a liberalized market in which prices are determined through the real-time interaction of supply and demand, Ecuador retains regulated final consumer tariffs. ARCONEL sets equitable tariffs to prevent abusive pricing and ensure universal accessibility, thereby providing a social counterweight to increased private investment.

Ecuadorian legislation has evolved from a rigidly state-centered approach toward an active state policy for attracting capital. The principal legal issue is no longer whether an electricity market exists, but how its regulatory mechanisms should be calibrated.

The current framework demonstrates that an electricity market can be both viable and constitutionally valid, provided that it operates under clear delegation rules and preserves state authority over tariffs and sector strategy.

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